«Spukhafte Fernwirkung»? Local Born geometry, the Tsirelson bound and the singlet curve: registered consistency checks, checkerboard interference and a retrocausal singlet model

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Abstract

Bell experiments without loopholes give CHSH values above the local bound 2, and high-precision measurements are compatible with Tsirelson's quantum bound 2√2 (Poh et al. 2015: 2.82759 ± 0.00051). This record asks what the author's framework, Mathematical Materiality (MM), must contain to respect both facts, and where it then stands among interpretations of quantum mechanics. No mathematical priority is claimed: the theorems used are published (Tsirelson; Barnum et al. 2010; Sorkin's quantum measures; Craig, Dowker et al. 2007; Aharonov's two-time formalism; Cramer; Argaman 2010; Wharton and Argaman 2020), and the record separates published results, derivations in the working notes and numerical conjectures. Predictions were registered with SHA-256 fingerprints before each test; failed predictions are reported. (1) Local readings and the Tsirelson bound. If each wing is read with the Born geometry (Euclidean state space with all its effects), no no-signalling joint event exceeds 2√2. Replacing the Euclidean ball with ℓ^p balls raises the maximum: 2^(2−1/p) for p ≥ 2 (the value of Ver Steeg and Wehner 2009), above 2√2 also for 1 < p < 2 (exact value conjectural). With marginals of length m the maximum is the quantum 2√(2 − m²). Two registered predictions failed: a hand formula, wrong for p < 2, and a three-dimensional search that stayed below the value already reached by the two-dimensional construction.

«Spukhafte Fernwirkung»? How strongly can two distant particles be linked, and why quantum probabilities are a square

Student version of MM record 2.48.2 (DOI 10.5281/zenodo.23777520), written with an AI assistant from the archived text; the archived text remains on Zenodo. The archived text is in English.

The title quotes Einstein. «Spukhafte Fernwirkung» in German means "spooky action at a distance". Einstein wrote it in a letter to Max Born of 3 March 1947, speaking of the linked particles of quantum mechanics. The word «Verschränkung», that is, entanglement, the quantum link, is instead Schrödinger's (1935).

The versions of this work

- **Version 2.48, 13 October 2026.** It is the starting point. It continues MM 2.40, the Feynman checkerboard (DOI 10.5281/zenodo.23280168). It also updates where MM stands among the interpretations of quantum mechanics (section 5). And it corrects texts on quantum foundations published in versions 2.7–2.27 of *Mathematical Materiality: a status report* (concept DOI 10.5281/zenodo.21802475; section 6).
- **Version 2.48.1, 26 October 2026.** It adds section 4.6: Born probabilities from counting copies of equal modulus. It also corrects the sentences of sections 1, 2, 7, 8, 9 and 10 that the new section makes inaccurate or incomplete. Programs, results and registrations of blocks 01 to R3 stay the same.
- **Version 2.48.2, 29 October 2026.** It adds section 4.7: the singlet correlation −cos θ from the two ends of the nexus. It puts in the new title with Einstein's phrase. It adds rows to sections 2, 7, 8 and 9. Everything else is the text of 2.48.1. Programs, results and registrations of the earlier blocks stay the same.

1. What this work is, and what it is not

A link in the MM programme

This work is **a link** in the MM programme: the one that concerns quantum mechanics. It shows how four working assumptions fit together (section 3). For each one it reports numerical consistency checks, registered beforehand. And it discusses their relation to published experimental results. The four assumptions are:

- local Born readings;
- weights built from pairwise correlations;
- a future boundary that selects the outcome, a "loop";
- the imaginary weight of the turn, on the checkerboard of MM 2.40.

We will make these words clear in section 3.

What it is not

It is **not** a derivation of quantum mechanics. And it does not distinguish MM from standard quantum mechanics with an experiment.

For the Born rule there is one more result, in section 4.6 (new in 2.48.1). In the 2.40 model the count of copies of equal modulus gives $|\psi|^2$, for rational ratios, that is, fractions. Three declared assumptions are needed: H-es, H-eq and H-add. For irrational ratios H-cont is also needed. The calculations cover denominators up to 12. For arbitrary fractions the general construction of the cited sources is used.

No mathematical priority is claimed: the text does not say that it was the first to discover these mathematical results.

Three kinds of statement kept apart

- results published by others (section 2);
- derivations written in the working notes; several were supplied by the verifier;
- numerical conjectures, that is, results found on the computer but not proved.

Some predictions are **registered**: written in a file and fixed with a fingerprint before the tests. So they cannot be changed afterwards. The results found afterwards are marked as such. **The predictions that failed are reported as failed** (sections 4.1, 4.2, 4.3). Each block, and this text, was checked by an independent artificial intelligence system.

2. Prior work, first

An idea to understand: the score S

Before the table, an idea that is needed throughout the text. Let us take two particles born together, which then move apart. Two experimenters, usually called Alice and Bob, measure them, each on their own side. Each can choose between two ways of measuring. Each measurement gives +1 or −1. With many trials one calculates a score, **S**, which says how much the results of the two sides agree. It is the **CHSH** combination, from the names of Clauser, Horne, Shimony and Holt.

- If the two particles carry with them a common "instruction sheet", written at birth, S does not exceed **2**. It is the Bell limit, the **local bound**.
- Quantum mechanics goes up to $\mathbf{2\sqrt{2}}$, about 2.83. It is the **Tsirelson bound**.
- A theory that forbids only sending messages faster than light could go up to **4**. It is the **no-signalling** maximum (the "PR boxes").

Experiments find values above 2 and up to about 2.83. The basic question is: why exactly $2\sqrt{2}$, and not 4?

The table of prior work

topic	published result used here
local bound	Bell 1964; Clauser, Horne, Shimony, Holt, PRL 23, 880 (1969); Fine, PRL 48, 291 (1982)

quantum bound	Tsirelson (1980): 2√2
no-signalling maximum	Popescu, Rohrlich, Found. Phys. 24, 379 (1994): S = 4
bipartite: full local quantum structure + no signalling ⇒ quantum correlations	Barnum, Beigi, Boixo, Elliott, Wehner, PRL 104, 140401 (2010)
local state-space geometry and non-locality	Janotta, Gogolin, Barrett, Brunner, NJP 13, 063024 (2011); Ver Steeg, Wehner, Quantum Inf. Comput. 9, 801 (2009), where the value 2^(2−1/p) already appears in their (different) p-norm theories; Bluhm, Jenčová, Nechita, CMP 393 (2022); Loulidi, Nechita, PRX Quantum 3, 040325 (2022)
tools	Clarkson, Trans. AMS 40, 396 (1936); Riesz, Acta Math. 49, 465 (1927) and Thorin, Kungl. Fysiogr. Sällsk. i Lund Förh. 8, 166 (1939) (Riesz–Thorin theorem); Horodecki R., P., M., Phys. Lett. A 200, 340 (1995)
quantum measure (grade 2)	Sorkin, Mod. Phys. Lett. A 9, 3119 (1994)
Born rule from envariance and counting (2.48.1, section 4.6)	Zurek, PRL 90, 120404 (2003); PRA 71, 052105 (2005); Phil. Trans. R. Soc. A 376, 20180107 (2018); objections: Schlosshauer, Fine, Found. Phys. 35, 197 (2005); Barnum, quant-ph/0312150 (2003); Mohrhoff, Int. J. Quantum Inf. 2, 221 (2004)
decision-theoretic and self-locating derivations (2.48.1)	Deutsch, quant-ph/9906015 (1999); Wallace, quant-ph/0312157 (2003) and arXiv:0906.2718 (2009); Saunders, Proc. R. Soc. A 460 (2004); Sebens, Carroll, Br. J. Philos. Sci. 69, 25 (2018); critical analysis of Deutsch's derivation: Barnum, Caves, Finkelstein, Fuchs, Schack, Proc. R. Soc. A 456, 1175 (2000)
Born rule from structure (2.48.1)	Busch, PRL 91, 120403 (2003); Masanes, Galley, Müller, Nat. Commun. 10, 1361 (2019)
Born square for radiative processes, transactional (2.48.1)	Kastner, Cramer, arXiv:1711.04501 (2018)
local model with a correlation linear in the angle (2.48.2, section 4.7)	Bell, Physics 1, 195 (1964)
cos² law from statistical distinguishability or information (2.48.2)	Wootters, PRD 23, 357 (1981) and Entropy 15, 3130 (2013), arXiv:1301.2018; Summhammer, quant-ph/9910039; Brukner, Zeilinger, Found. Phys. 39, 677 (2009), arXiv:0905.0653
binary alternative, SU(2) and three-dimensional space (2.48.2)	von Weizsäcker, Aufbau der Physik (1985), reviewed by Lyre, quant-ph/0309183; Müller, Masanes, NJP 15, 053040 (2013)
spinors and null directions (2.48.2)	Penrose, Rindler, Spinors and Space-Time, vol. 1 (1984), cited from memory
joint quantum measures and Tsirelson	Craig, Dowker, Henson, Major, Rideout, Sorkin, J. Phys. A 40, 501 (2007): **strong** positivity gives Tsirelson; Barnett, Dowker, Rideout, J. Phys. A 40, 7255 (2007): without it PR boxes are allowed; Dowker, Henson, Wallden, NJP 16, 033033 (2014)
Hilbert space from a strongly positive decoherence functional	Dowker, Johnston, Sorkin, arXiv:1002.0589 (2010); it does not by itself give the dynamics, composite systems or frequencies
reconstructions	Chiribella, D'Ariano, Perinotti, PRA 84, 012311 (2011): purification together with other axioms
checkerboard and Dirac walk	Feynman, Hibbs (1965); Bisio, D'Ariano, Tosini (2015); MM 2.40
two-time formalism	Aharonov, Bergmann, Lebowitz, Phys. Rev. 134, B1410 (1964); Aharonov, Gruss, quant-ph/0507269 (2005); Aharonov, Cohen, Gruss, Landsberger, Quantum Stud. Math. Found. 1, 133 (2014); Gell-Mann, Hartle, gr-qc/9304023 (1994)
transactional interpretation	Cramer, Rev. Mod. Phys. 58, 647 (1986); Kastner, *The Transactional Interpretation of Quantum Mechanics* (Cambridge, 2012); Maudlin's contingent-absorber objection, *Quantum Non-Locality and Relativity*, 3rd ed. (Wiley-Blackwell, 2011)
retrocausal, locally mediated models	Costa de Beauregard, C. R. Acad. Sci. 236, 1632 (1953); Argaman, Am. J. Phys. 78, 1007 (2010); Price, arXiv:1002.0906 (2012); Leifer, Pusey, Proc. R. Soc. A 473, 20160607 (2017); Wharton, Argaman, Rev. Mod. Phys. 92, 021002 (2020); Almada et al., arXiv:1510.03706 (2016)

fine-tuning of no-signalling	Wood, Spekkens, New J. Phys. 17, 033002 (2015)
non-equilibrium and signalling	Valentini, Phys. Lett. A 156, 5 and 158, 1 (1991)
Local Friendliness	Bong et al., Nature Physics 16, 1199 (2020); Cavalcanti, Wiseman, arXiv:2106.04065 (2021)
self-consistency	Friedman, Morris, Novikov, Echeverria, Klinkhammer, Thorne, Yurtsever, Phys. Rev. D 42, 1915 (1990)

Published measurements and benchmarks

- **CHSH**, that is, the measured score S: Poh and colleagues, PRL 115, 180408 (2015): S = 2.82759 ± 0.00051; Hensen and colleagues, Nature 526, 682 (2015): 2.42 ± 0.20; Storz and colleagues, Nature 617, 265 (2023): 2.0747 ± 0.0033.
- **Third-order interference.** With three slits, quantum mechanics says that there is nothing new beyond the pairs of slits. Sinha and colleagues, Science 329, 418 (2010): below 10^{-2} of the pairwise term. Kauten and colleagues, NJP 19, 033017 (2017): more than four orders of magnitude below.
- **Quantum theory with real or complex numbers.** Renou and colleagues, Nature 600, 625 (2021) (theory). Li and colleagues, PRL 128, 040402 (2022): more than 4.5 standard deviations, in a network with two independent sources.
- **Diósi–Penrose collapse:** Donadi and colleagues, Nature Physics 17, 74 (2021).

3. The working assumptions

The assumptions come from the author of the framework as premises. His exact words are kept outside this deposit. Here they are written in technical form, with what the translation adds.

First, two basic words. In quantum mechanics every possible outcome has an **amplitude**. Think of it as an arrow: it has a length and a direction. The **Born rule** says that the probability of an outcome is the square of the length of the arrow: $|\psi|^2$. An example with a number chosen by us: an arrow of length 0.6 gives probability 0.36. The **singlet** is a state of two linked particles: if the spin of the two is measured in the same direction, the results are always opposite.

- **W0 – local Born reading.** Each measured system is read with the Born rule of a two-level system, like a quantum coin. Its states form a Euclidean ball: the Bloch ball (d = 3) or its real disc (d = 2), with all its dual effects. "Euclidean" means that distances are measured as in everyday geometry, with Pythagoras' theorem. As a control, section 4.1 replaces the Euclidean ball with real ℓ^p balls. In these, distances are measured in another way: state $\|r\|_p \le 1$, effects $\|u\|_q \le 1$, $1/p + 1/q = 1$. With p = 2 the normal ball is recovered. B(p) denotes the maximum value of S for d = 2.
- **W1 – weights from copies correlated in pairs.** The weight of an event, that is, of a set of "copies", depends only on the single copies and on the pairs: $\mu(A) = \Sigma_{\alpha,\beta\in A} D_{\alpha\beta}$. It is a grade-2 measure in Sorkin's sense. *Added in translation:* D is positive semidefinite (a Gram matrix), so $\mu(A) = \|\Sigma_{\alpha\in A} v_\alpha\|^2$. The premise suggests these two steps, but does not impose them.
- **W2 – the loop.** The outcome written in a record is selected by the future end of its history. The history closes on itself, like a ring. In section 4.3 this is a future boundary chosen at random with a certain measure. In section 4.4 it is a hidden parameter that depends on a future setting.
- **W3 – the imaginary weight of the turn.** On the checkerboard of 2.40 a particle moves forward in a zigzag. A turn has weight $-i \sin\theta$. A straight step has weight $\cos\theta$. Here θ is the mass per tick, and i is the imaginary number, the one with $i^2 = -1$.
- **A word not to be misunderstood.** In MM texts "non-local" means *not located in spacetime*, pregeometric. It concerns the deepest level of the framework. The events of experiments, the settings and the records are located in spacetime as usual. "Non-local" does not mean an influence faster than light.

4. Tests and results

4.1 Local readings and the Tsirelson bound

The question. Why does S stop at $2\sqrt{2}$? This section shows that the cap comes from the way each side is read, on its own.

The model. There are two wings, with the local spaces of W0. The joint event is any matrix C, with marginals, that gives probabilities that are never negative for every pair of local effects. This is called the maximal tensor product, without signalling. The **marginals** are the averages of the results of one side only, looked at on its own. To look for the

maximum of S without constraints, one can take the marginals to be zero and the observables unbiased, without losing anything. Then the constraint becomes $\|C\|_{q\to p} \le 1$. The case with marginals of fixed length m is treated separately below. H = [[1,1],[1,−1]].

The derivations (in the working notes, checked by the verifier).

- **Upper bound.** $S \le \|v_0+v_1\|_q + \|v_0-v_1\|_q \le 2^{(2-1/\max(p,q))}$, by Clarkson's inequalities, in any dimension.
- **p = 2.** The bound is 2√2 for every admissible joint event. It comes from the parallelogram law: it is Tsirelson's proof.
- **p ≥ 2.** The bound is attained by $C = 2^{(-1/p)}H$ (Riesz–Thorin with complex endpoints). So $B(p) = 2^{(2-1/p)}$.
- **1 < p < 2.** B(p) > 2√2 (the proof was supplied by the verifier). The upper bound is not attained. The best value found is the Hadamard construction $2^{(1+2/p)}/\|H\|_{q\to p}$. That it is the maximum is a **conjecture**.
- **Marginals of length m (p = 2, d = 2).** The maximum is exactly $2\sqrt{(2 - m^2)}$, the quantum value (proof supplied by the verifier). S = 2√2 forces both marginals to be zero.

The maximum of S for d = 2. It is proved for p ≥ 2 and at the endpoints. For 1 < p < 2 these are the best values found on the computer.

p	**1**	**1.1**	**1.25**	**1.5**	**1.75**	**1.9**	**2**	**2.1**	**2.5**	**3**	**4**	**8**	∞
value	4	3.6526	3.3726	3.1065	2.9428	2.8699	2.8284	2.8755	3.0314	3.1748	3.3636	3.6680	4

Look at the table: the lowest value is exactly at p = 2, where it is 2.8284, that is, 2√2. For every other p the maximum is higher.

Registered outcomes.

- **Passed:**
- a common record read locally gives at most 2 (T1);
- 2√2 at p = 2, in d = 2 and 3 (T2);
- 4 at p = 1 and ∞ (T3);
- marginals (T6);
- only p = 2 allows continuous rotations that preserve the norm (T7: distortion ≥ $1.1\cdot10^{-2}$ for p = 1.5, 3, 4).
- **T4 failed for p < 2.** The maxima found on the computer lie from 0.37 to 3.15 % below the hand formula $2\sqrt{2}\cdot2^{|1/p-1/2|}$, registered before the calculation. The error of the hand calculation: it used the real value $\|H\|_{\infty\to 1} = 2$ in a complex Riesz–Thorin interpolation, where the right value, the complex one, is 2√2.
- **T5 failed as written.** The random search in d = 3 stayed below the d = 2 value. At p = 3 it fell short by $4.8\cdot10^{-4}$, outside the tolerance 10^{-4}: this case fails, and with it the test. At p = 1.5 it fell short by $9.4\cdot10^{-5}$ in relative terms (inside the tolerance) or by $2.9\cdot10^{-4}$ in absolute terms (outside). At p = ∞ it matched.
- The registered tolerance did not say whether it was relative or absolute. The program used the relative one.
- The search did not try the known construction without changing it.
- The second round, R2-A, shows that the construction placed in d = 3 reproduces the d = 2 value to $3\cdot10^{-16}$. No candidate in d = 3 exceeded it. This, however, does not prove that for p < 2 the result does not depend on the dimension.

What it means. Within the declared model (two sides, maximal tensor product), each side read with the Euclidean geometry (with all its dual effects) gives |S| ≤ 2√2. **The cap comes from the local geometry, not from what links the two wings.** The measured value, S = 2.82759 ± 0.00051, is compatible. These data, however, do not measure p. An ℓ^p reading would allow S above 2√2, but would not require it.

4.2 Two slits on the checkerboard

The model. The Dirac walk of 2.40 is used, with θ = 0.5. There are two slits at ±20 and a screen 1000 steps later. Three "coins" are tried, that is, three rules for the steps:

- the coin of 2.40, with $-i \sin\theta$ on the turns;
- a real rotation coin, without imaginary numbers;
- a coin with positive weights only.

Registered outcomes.

- **First run, passed:**

- positive amplitudes never give a negative cross term;
- the real coin reproduces the probabilities of 2.40 to $3.3 \cdot 10^{-15}$, for sources in one direction and for their mixture (G = diag(1, i) takes one coin into the other);
- the third-order term $|I_3|/\max|I_2| = 8 \cdot 10^{-16}$ for the quadratic reading. In this statistic the variants |ψ|^p with p = 1 and 3 give 0.80 and 0.39. It is not the normalisation used by Sinha and colleagues;
- the matrix of the cross-correlations is positive (by construction).
- **First run, passed with limits:**
- T3b compares the same vector of numbers in the two models, not corresponding preparations: it gives no information;
- T5b compares the minimum fringe ratio over different regions.
- **First run, not met:** the dark fringes (T2) and the fringe spacing (T6). The registered "central region" contained only 2 sites of the lattice. The reason: the weight of a single slit has its peak on the ballistic front, that is, on the two fronts that move away from the slit at constant speed, not in the middle. **The definition was wrong, not the code.**
- **Second run**, registered after seeing the first, with the two slits lit in the same way (all passed):
- minimum ratio $\mu_AB/(\mu_A + \mu_B) = 4.1 \cdot 10^{-4}$;
- dark fringes at −143.2 and +140.4 sites (lattice sites −144 and 140), against ±142.1 predicted by the stationary phase with the lattice dispersion cos ω = cos θ cos k;
- separation 283.6 against 284.3;
- with positive weights the ratio never goes below 1.000000.

What it means.

- On the checkerboard every phase comes from the turns. At equal length, two more turns multiply the amplitude by $-\tan^2\theta$. The minus sign makes two contributions to the same outcome cancel each other: it is **destructive interference**. This is how the dark fringes arise. **Amplitudes that are only positive cannot produce a dark fringe.** Think of two water waves: if a crest meets a trough, the water stays flat. With only crests, the water never stays flat.
- For the preparations tested here, an equivalent real coin reproduces the probabilities. General comparisons require transforming the preparations and the measurements in the same way too.
- Li and colleagues distinguish real quantum theory from complex quantum theory in a network with independent sources, under the composition assumptions of that test (that is, on how systems are combined). These calculations do not address that question.

A declared defect. The diagnosis script imported the first program and rewrote its results file. The physical values are identical to the registered log. Only the running time changed. The second round, R2-C, does the same diagnosis in a temporary folder and gives the same output.

4.3 Can a future boundary give Born frequencies?

The idea. Here the assumption W2, the loop, is tested. Imagine that the outcome of a measurement is chosen not only by the past, but also by how the history "closes" in the future. Are the frequencies obtained those of the Born rule?

The model.

- The "forward" state on the screen comes from the checkerboard, with the screen divided into 8 intervals.
- The "backward" walker uses the adjoint step U† = C†S†. Its coin has +i sin θ on the turns.
- The future is a minimal space made of record + environment. The rules for choosing the future are compared by their distance from the Born weights w_b. The distance used is the **total variation distance** (TVD): 0 means equal, 1 completely different. The w_b are conditioned on arrival in [−900, 900] and renormalised over the 8 intervals, like the frequencies.

Registered outcomes (all numerical thresholds met). Forward × backward is equal to the Born weight to $3 \cdot 10^{-16}$. But this is a mathematical identity, not a physical choice.

rule for the future	TVD from Born
uniform (Haar) future weighted with the loop closure $w_b \cdot \lvert\Phi_b\rvert^2$	0.0010
futures with a definite record, prior ∝ number of states d_b, weighted with the closure	0.0069

uniform future, unweighted, outcome by the two-state rule (ABL)	0.099
uniform future, unweighted, the largest closure wins	0.164
definite records, each record equally likely	0.489
definite records weighted with the closure, each record equally likely (different d_b)	0.316
K independent copies of the record, the largest closure wins (K = 1, 2, 4, 8, 16)	0.164, 0.042, 0.178, 0.285, 0.353

- The last row does not always grow with K: it goes up and down. The registered criterion (K = 16 farther than K = 1, K = 8 above 0.1) is met. **The word "grows" in the prediction is not.**
- **Deviation from the protocol.** The first run widened the two outer intervals of the screen. The values in the table come from the run with the registered intervals. 0.74 % of the weight falls outside the screen.

Found after the run, not registered: the exponential race.

- Take independent exponential random variables with the same rate, E_b. Choosing $\operatorname{argmin}_b(E_b/w_b)$ draws branch b with probability $w_b/\Sigma_c w_c$. It is like a race: the time of each branch is divided by its weight, and the smallest time wins. So a branch with more weight wins more often.
- The squared components $|\Phi_b|^2$ of a Haar-random complex vector (with dimension ≥ number of branches) are variables of this kind, divided by a common sum. So the rule "smallest $|\Phi_b|^2/w_b$ wins" reproduces Born (TVD 0.0024).
- It is an exact sampling algorithm. But it already contains the weights.

What it means. In this minimal model a future boundary gives the Born frequencies with some rules (the one weighted with the closure, or the exponential race), and not with others. **The measure over futures remains a postulate.** In the rule with prior $\propto d_b$ the factor d_b cancels against the mean overlap $1/d_b$.

4.4 A singlet model in which the future acts on the past, along local paths

The idea. In the singlet the two particles give linked results, even if they are far apart. Can one build a model in which nothing goes faster than light, but the future acts on the past? The model of this section adapts to spin ½ a model by Argaman (2010). It is called **retrocausal**: a cause in the future, an effect in the past.

The model.

- The source is at the point (0, 0) of the checkerboard. The readings are at $(\mp L, T)$. The choices of the settings are made at time $T - 10$. They are "space-like" separated from the reading of the other wing: no signal at the speed of light can go from one to the other in time. The checkerboard gives only the shape of the paths. The law that depends on the angle is assumed.
- Alice's future outcome fixes a parameter $\lambda \in \{a, a + \pi\}$, with probability p. The parameter is carried back along Alice's path to the source. Then it goes forward along Bob's path, as $\lambda + \pi$.
- Bob reads it with the local Born rule $\cos^2((\lambda_B - b)/2)$.
- In exact form, with $m = 2p - 1$ and $c = \cos(a - b)$: $P(A = u, B = v) = (1 + um)(1 - uvc)/4$. So $E = -c$ for every p, $\langle A \rangle = m$ and $\langle B \rangle = -mc$.
- At p = ½ these are exactly the singlet probabilities for projective measurements in the same plane.
- For other values of p, E does not change. But the marginals allow signals to be sent.

Registered outcomes (all passed).

- E within 0.0042 of $-\cos(a - b)$ on a 12×12 grid (threshold 0.008), and $S = -2.8281$.
- No observable signal at p = ½: 0.0039 and 0.0040 (threshold 0.008).
- Signal equal to $|2p - 1|$ for p = 0.6, 0.75, 0.9: 0.202, 0.502, 0.801 (tolerance 0.01).
- A "forward"-only control, with sign responses, gives $|S| = 2.0000$.
- The check of the shape of the paths is passed (with L = T the paths are straight).
- The symmetric mixture, in which the future of one wing or the other chooses, coincides with $-\cos(a - b)$ within 0.0037.

Second round R2-B (all passed).

- Born responses with λ uniform and independent of the settings: $E = -\frac{1}{2}\cos(a - b)$ within 0.0055, $|S| = 1.4157$.
- Real zigzags with $T > L$: λ stays constant along every path in 20,000 histories out of 20,000.
- Screening: with (λ_B, b) fixed, Bob's frequency does not depend on (a, A). The maximum is 1.58 standard errors over 8 comparisons, with threshold 5.
- Symmetric versus asymmetric version, compared directly: within 0.0046 (threshold 0.012).

What it means.

- With a dependence on future inputs, $\rho(\lambda \mid a, b) \neq \rho(\lambda)$, and a local Born reading, the singlet correlations are reproduced. Every link of the chain runs along paths inside the light cones, through the common past of the two readings.
- In this family of models signals cannot be sent **if and only if** the future chooses with the balanced weight, the Born one. This link is already discussed by Argaman (2010). And the $\cos^2$ law of the angle is assumed.
- The cancellation at $p = \frac{1}{2}$ is a "fine-tuning" in the sense of Wood and Spekkens: a precise value that makes an effect vanish.
- The parameter must not be readable at the source. Read there, it would reveal a, up to π.
- What it covers: one singlet pair, projective measurements in the same plane, ideal detectors.

4.5 Self-consistency and the balanced weight

The question. In section 4.4 the balanced weight $p = \frac{1}{2}$ is put in by hand. Why exactly that weight? The author's premise answers that it is not a "why" question: it is the existence of the present that chooses its past. Put technically: only **self-consistent** histories exist (Novikov; Friedman and colleagues 1990).

Think of the famous grandfather paradox. If you travel into the past and prevent your own birth, who set out? Novikov says: only histories without contradictions can exist.

The calculation (binomial distribution computed numerically, without random draws).

- In the model of 4.4 with $p > \frac{1}{2}$, Alice encodes a bit x in her setting (a = 0 or π, b = 0). A bit is a piece of yes/no information.
- N is odd. The N pairs are assumed independent and identical, given x. Then Bob's frequency of B = +1 is $\frac{1}{2} \mp \varepsilon/2$, with $\varepsilon = |2p - 1|$. Bob decodes x by majority. For $p < \frac{1}{2}$ the decoder must be inverted; the grid uses $p \geq \frac{1}{2}$.
- **An assumption, not calculated.** A faster-than-light channel is assumed which, combined with a second one in another frame of reference, brings the decoded bit back into Alice's past, without other errors. It is Tolman's "antitelephone" (Tolman, *The Theory of the Relativity of Motion*, 1917; Benford, Book, Newcomb, Phys. Rev. D 2, 263, 1970). The pair model does not provide this construction.
- **The paradox protocol.** Alice sets x = NOT(decoded bit): she does the opposite of what she receives. A history is consistent only if Bob decoded NOT x.
- W(N, p) = P(decoded bit = NOT x) is the weight of the consistent histories, calculated *before* imposing consistency.

p	N = 1	11	101	1001	10001
0.50 (Born)	0.5	0.5	0.5	0.5	0.5
0.51	0.490	0.473	0.420	0.263	0.0227
0.55	0.450	0.367	0.156	$7.6 \cdot 10^{-4}$	$5.9 \cdot 10^{-24}$
0.60	0.400	0.247	0.0209	$8.1 \cdot 10^{-11}$	$4.3 \cdot 10^{-91}$
0.75	0.250	0.034	$3.3 \cdot 10^{-8}$	$6.4 \cdot 10^{-65}$	below the smallest computable value (positive, below double precision)

Registered outcomes (all passed).

- S1: with $p = \frac{1}{2}$, $W = \frac{1}{2}$ for every N of the grid.
- S2: for the values $p \neq \frac{1}{2}$ tested, W decreases with N, below the registered bounds.
- S3: for every $p \neq \frac{1}{2}$ tested, some N of the grid gives $W < 0.05$.

A general statement, proved afterwards with Hoeffding's inequality (supplied by the verifier): $W(N, p) \leq \exp(-N\varepsilon^2/2)$. So

$W \to 0$ for every fixed $p \neq ½$.

What it shows, and what it does not.

- **Shown.** With any weight other than Born's, a protocol with enough pairs makes the prior weight of the consistent histories (the weight before conditioning on consistency) as small as one likes. With the Born weight the channel carries no information. So with the Born weight the antitelephone cannot be built at all, and $W = ½$ is only a formal comparison.
- **Not shown.** That a weight other than Born's is *impossible*. For $0 < p < 1$ and every finite N, $W > 0$: the consistent histories always have a non-zero weight. (At $p = 0$ or 1, $W = 0$ already with one pair.)
- **What is needed to pass from "arbitrarily improbable" to "cannot exist".** One more postulate is needed. A formulation close to the author's premise is: *a present exists only if no protocol can make its consistency arbitrarily improbable*. Here it is checked only for the family of protocols described (inf over N of $W > 0$). It is assumed that the return link and the resources are available for N as large as one likes. It is written as a **postulate of the author**, not derived.
- **Another way to impose consistency: conditioning.** When $W > 0$, $P_C(h) = P(h) \cdot 1_C(h)/W$ gives the consistent histories total probability one. But it changes the statistics. The post-selected closed timelike curves of Lloyd and colleagues (PRL 106, 040403, 2011) exclude the histories with zero post-selection amplitude and renormalise the others. Which measure describes the closed loop is part of the postulate.
- **What it covers:** one protocol, within the pair model. The local $\cos^2$ law is assumed throughout.
- **Who had already said it.** Signals outside the Born weights are known for pilot waves (Valentini, Phys. Lett. A 158, 1, 1991) and for this toy model (Argaman 2010). The reading "existence of the present ⇒ balanced weight" is the author's.
- **Registration.** The prediction file was written before the code. Both got their fingerprint in the same minute, before the run.

4.6 Born probabilities by counting copies of equal modulus (new in 2.48.1)

The question. With which declared assumptions does the counting of copies, in the 2.40 model, give the probability $p = |\psi|^2$? And where does the square enter?

The author's premises. Three sentences of the author are kept in Italian, with his exact words, in a file of the package. The archived text gives them translated into English; here they are in that translation:

- U1 (23 October 2026): if you have to describe a 10 by means of a number 1 (10 = 1), you are forced to distribute the descriptor 1 over 10 space-times of the number 10;
- U2 (13 October 2026): the rule is decided by the number and the mutual correlation of the copies of the nexus;
- U3 (13 October 2026): it is the future of a history that chooses its past, so the answer is the loop.

Their translations into mathematical objects are **choices of the AI** (the "bridges" P1–P6 of the protocol), not words of the author.

Prior work, first. Twenty-two sources were read with the same eight questions: 19 downloaded from arXiv, 3 already in the working records. The derivations that count copies (Zurek 2003, 2005, 2018; the ancilla step of Deutsch 1999; Saunders 2004; Sebens and Carroll 2018) have three ingredients in common:

1. *equal modulus, equal probability.* It is the step that the critics attack. For Schlosshauer and Fine (2005) it is an additional assumption about the connection between state and probabilities. For Barnum, Caves, Finkelstein, Fuchs and Schack (2000) Deutsch's version hides a principle of insufficient reason. Envariance gives a symmetry; equal probabilities need a further, contested link;
2. *dividing into copies of equal modulus, and counting.* An amplitude $\sqrt{(m/M)}$ becomes m copies of modulus $1/\sqrt{M}$, and the probability is m/M. None of the critics contests this step;
3. *the square comes from the conserved norm*, not from the count. Zurek attributes it to Pythagoras, that is, to the Euclidean geometry of Hilbert space. Saunders shows that with an evolution that conserves an l^p norm the same argument would give |c|^p.

An example, with the numbers of test K2 (in the table further below). An amplitude $\sqrt{(3/4)}$ is divided into 3 copies, each of modulus $1/\sqrt{4} = ½$. The amplitude that remains, $\sqrt{(1/4)}$, is 1 equal copy. Out of 4 equal copies, 3 belong to the first outcome: probability 3/4. And it is exactly the square of $\sqrt{(3/4)}$.

Other routes:

- the decision-theoretic derivations (Deutsch; Wallace 2007, 2009) use other axioms: branching indifference, dependence on the state alone, continuity;
- the structural routes do not count branches: Gleason and Busch (2003), Masanes, Galley and Müller (2019), Sorkin (1994);
- in the ABL rule of the two-state vector formalism, probability as squared amplitude is part of the rule;
- Kastner and Cramer (2018) present a derivation of the Born square for radiative processes, in direct-action theory. They start from the requirement that a real photon involves both emission and absorption (their pp. 1 and 8-9). That the absorber's response carries the dual vector, and that the product is a probability, rests on the transactional picture (p. 2 and note 1). Both arguments are different from the copy counting of B-2.

No source derives that probabilities exist, or which outcome occurs. The data: three-path interference is zero within $|\kappa| < 0.01$ (Sinha and colleagues 2010) and $\kappa_3^{\sim} = 0.0 \pm 3.1 \cdot 10^{-5}$ in the classical regime of Kauten and colleagues (2017).

The protocol. Five formulations:

- B-1: counting the paths without phases (a control);
- B-2: counting copies of equal modulus (from premise U1);
- B-3: number and mutual correlation (U2);
- B-4: the loop (U3);
- B-5: which norm the coin conserves.

With named assumptions:

- H-es: probabilities exist and depend only on the state of the copies;
- H-eq: copies of equal modulus are equally probable;
- H-add: the probability of a set of distinct copies is the sum, normalised to 1;
- H-cont: continuity for irrational ratios;
- H-prod (B-4 only): the product of the forward and backward amplitudes is a probability.

The protocol was checked twice by a second AI system, with 9 and 4 remarks. All were accepted in two corrections registered before any calculation. There was no third round. The main points of the verifier became part of the classification: B-3 defines its measure as $|\Sigma w|^2$, which is already the Born form; B-4 uses a quadratic rule as a probability; the complex conjugate of the backward amplitude follows from defining the return with the adjoint step.

Tests and results. The program was registered before the run. But the expected outcomes were written knowing the physics of quantum walks. **So they are not blind predictions.**

test	what it measures	result
K1, 2.40 coin	two slits, r = I_AB/(I_A + I_B), geometry of 4.2	min r = $4.1 \cdot 10^{-4}$ (dark fringes; criterion min r < 0.05)
K1, paths without phases	the same statistic with the path counts	r = 1 at every site (to $2 \cdot 10^{-16}$): no dark fringe
K2	one coin step at $\theta = \pi/6$, division into 4 cells	4 cells of modulus ½; counting gives 3/4 and 1/4 = $\|\psi\|^2$
K2b	every ratio m/M with M ≤ 12 (66 cases)	M cells of modulus $1/\sqrt{M}$; counting = $\|\psi\|^2$ to $9 \cdot 10^{-16}$
K3	variation of $\Sigma\|\psi\|^p$ over 200 steps, $\theta = \pi/6$, $\pi/4$, $\pi/3$, 0.3	p = 2: ≤ $2.7 \cdot 10^{-14}$; p = 1: 10.6–12.7; p = 1.5: 1.9–2.5; p = 3: 0.78–0.90; p = 4: 0.92–0.99
K4	Sorkin's κ with three slits at −20, 0, +20	$\kappa = 2.6 \cdot 10^{-17}$ at the central maximum; inside the windows of both experiments
K5	backward amplitude from the record	equal to the complex conjugate
K5bis (added after the run, declared)	the same at five sites with imaginary amplitude	distance from the conjugate ≤ $8 \cdot 10^{-17}$; from the amplitude itself 0.14–0.19

In the registered K5 the record site (the maximum of the amplitude) had a real amplitude. There "conjugate" and "equal" coincide. For this reason K5bis repeats the check where they are different.

Classification.

formulation	outcome	where the square enters
B-1 paths without phases	does not follow from the data (no dark fringe)	nowhere: the rule is linear
B-2 copies of equal modulus	derived with H-es, H-eq, H-add in the cases tested (ratios with denominator up to 12); for every rational ratio the general construction of the sources applies (not redone here), and H-cont for the irrational ones	in the number of cells, $m_k \propto \lvert\psi_k\rvert^2$, because the coin conserves the sum of squares
B-3 number and correlation	assumes the quadratic measure (as Sorkin; K4 checks the program)	in the definition $\mu(A) = \lvert\Sigma w\rvert^2$
B-4 the loop	circular for the Born rule; it remains a premise on the choice of the outcome	assumed: forward × backward is $\lvert\psi\rvert^2$ by the adjoint
B-5 the conserved norm	among the exponents, angles and state tested, only p = 2 stays constant	this is the point: the coin conserves the sum of squares and not the others

What it means, as a reading.

- In the 2.40 model the count of copies of equal modulus gives $|\psi|^2$ for rational ratios, with H-es, H-eq and H-add. Irrational ratios also require H-cont. The calculations cover denominators up to 12. For arbitrary fractions the general construction of the sources is used. The construction (dividing and counting) is similar to Zurek's. Here, however, H-eq is declared as a premise, not derived from envariance. And the number of cells is read from the amplitudes.
- The square comes from the 2.40 coin, together with the structure the model adopts: complex amplitudes, linear superposition, a coin that preserves the inner product. The same exponent 2 appears in two separate results of section 4.1. In the declared model the $2\sqrt{2}$ cap is proved for p = 2. And T7, on the rotations that preserve the norm, reports distortion for p = 1.5, 3 and 4.
- H-eq is the step contested in known physics. In the framework it is the author's premise U1 (bridge P3, a choice of the AI). **The objection does not disappear: it becomes a postulate.**
- The 2.40 coin is the Feynman checkerboard: known physics adopted as a postulate.

What is not claimed. That probabilities exist (H-es); which outcome occurs; a solution of the measurement problem. The tests are finite: one coin step for K2 and K2b; four angles and five exponents for K3. The general construction for arbitrary fractions is that of the sources.

4.7 The singlet curve $-\cos\theta$ from the two ends of the nexus (new in 2.48.2)

The question mark in the title. It is the point of this section. In the translations that hold, the "mirror" between the two particles is a reading rule of a single nexus with two ends. It is not a signal that travels between the two ends. And the marginals stay zero: nothing can be sent with the mirror. So no "spooky action at a distance".

The question. Sections 4.1–4.6 put caps on the correlations: 2 for a common record read locally, $2\sqrt{2}$ for Born readings at the two ends. But they never derive the curve that Bell experiments measure. It is the curve of two spin-½ particles in the singlet state: $E(a, b) = -a\cdot b = -\cos\theta$. Here θ is the angle between the two measurement directions. With the Born rule, a two-dimensional complex state space and rotations that act as SU(2), the curve follows in one line. The question is: which of these ingredients can come from the two ends of the nexus, and which stay imported? There are five ingredients:

- (i) the exponent 2;
- (ii) an end as a two-state system with a phase;
- (iii) the three directions of space as directions of the end, with the half angle;
- (iv) the singlet as mirror reciprocity of the two ends;
- (v) zero marginals, that is, no signal.

The method. Six translations, T0–T5, registered with the criteria C1–C5 before the program was run (13:24:31 UTC):

- C1: $E = -\cos\theta$ exactly, or within 10^{-3} on a grid;
- C2: marginals within 10^{-3} of zero;
- C3: $S = 2\sqrt{2}$ at the standard angles, and not above it on a grid of 64 directions;

- C4: dependence on the relative angle only, in the plane and for 200 random pairs of directions in space;
- C5: the list of what is imported.

The expected outcomes had been worked out by hand before the registration. **It is a declared contaminated preregistration, not a blind prediction.** The first registered version of the program stopped on a rounding error: a ratio of −1e-17 raised to the power ½. The corrected file, which keeps that ratio between 0 and 1, was registered before it was run (correction 1).

Results. Deviations written as 0 are below 10^{-15}.

translation	C1	C2	C3 (standard / grid maximum)	C4 plane / space	outcome	imported
T0 quantum mechanics (control)	0	0	2.828427 / 2.828427	0 / 0	holds, everything imported	(i)–(v)
T1 the 2.40 coin read by two oriented readers; 24 variants (k ∈ {½, 1}, phase χ ∈ {0, π/2, π}, weight |amplitude|^p with p = 1–4)	0 only for k = ½, χ = π, p = 2	0	2.828427 for that variant	0 / 0 for that variant	holds with imported ingredients (1 variant of 24)	p = 2; a two-component space with complex amplitudes and a tensor-product state; the sign χ = π; k = ½ and the azimuth; normalisation of the weights
T2 mirror: after the reading of A, the end B lies along −u·a and is read with a law f	—	0	—	—	does not determine the curve	—
T2 with f = (1 + c)/2	0	0	2.828427 / 2.828427	0 / 0	holds with the law of the end imported	f
T2 with f linear in the angle	0.2071	0	2 / 2	0 / 0	fails (Bell's straight line)	—
T2 with f = cos²(3θ/2)	1.5364	0	2.828427 / 2.828427	0 / 0	fails (−cos 3θ)	—
T3 copies counted with their noise (Wootters' route) + mirror	0	0	2.828427 / 2.828427	0 / 0	holds with imported ingredients	independent copies (binomial variance), linearity of mixtures, Fisher metric, equal distinguishability for equal rotations, end points, monotonicity, isotropy in 3D, fair first outcome, mirror
T4a three sides (2+1), probability = pair product normalised on the opposite direction	0	0	2.828427	0 / not defined	fails for non-coplanar settings (holds in the plane)	the pair rule, mirror
T4b four sides (3+1), same rule	0	0	2.828427 / 2.828427	0 / 0	holds with imported ingredients	the pair rule P = η(u_s, u_r)/η(u_s, u_−s) (translation by the AI), the fourth side, fair first outcome, mirror
T4 with q = ½ (probability (η/η_opp)^½)	0.1659	0.4142	2.164784	—	fails	—
T4 with q = 2	0	0.5	2.828427	—	fails on C2	—
T5 common record read locally (Bell 1964)	0.2071 on the grid, 0.2105 continuous	0	2 / 2	0 / 0	fails, as registered (control)	—

What the checks show.

- **T2, the mirror.** With P(A = u) = ½ and the mirror, E = 2f(−c) − 1 holds whenever f(c) + f(−c) = 1 (symbolic calculation). The mirror gives the perfect anticorrelation at θ = 0: measuring in the same direction, the results are always opposite. And it gives the form of the correlation. But it does not give the curve. With a law linear in the

angle it gives Bell's straight line and S = 2. **In this model, once the mirror is given, going beyond 2 depends on the law with which one end is read.** It is in line with section 4.1: the cap is set by the local reading.

- **T3, Wootters' route.** With $f = \cos^2 g$ the Fisher information is $4g'^2$. So "equal distinguishability for equal rotations" means g linear. Then $f(0) = 1$ and $f(\pi) = 0$ make κ a half-integer. Monotonicity selects $\kappa = ½$, that is, $\cos^2(\theta/2)$: the half angle, without putting SU(2) in by hand. If instead one counts the fractions without their noise, one gets the law linear in the angle (Bell's straight line). This is not new: Wootters (1980–81, not read directly here; known through Wootters 2013), Summhammer (quant-ph/9910039, equation 6), Brukner and Zeilinger (Found. Phys. 39, 677, 2009, with a quadratic measure of information assumed). It requires copies independent of each other. The author's premise of 13 October speaks instead of correlated copies.
- **T4, the light sides.** The forms $Q = \Sigma_{i<j} x_i x_j$ have eigenvalues $(1, −½, −½)$ for three sides and $(3/2, −½, −½, −½)$ for four: signatures (1, 2) and (1, 3). For null directions, that is, light directions, the normalised product gives $\sin^2(\theta/2)$ (deviation $4 \cdot 10^{-16}$). Writing a light direction as $2\psi\psi^T$ rotates ψ by $\theta/2$. With three sides the readings lie in a plane. With four sides they cover the three directions, and T4b gives $−a \cdot b$ in space. The calculation uses only real directions. Complex numbers are not needed for these correlations, and the fourth side does not impose them. They enter only if one wants to write a light direction as the product of a two-component object ($S^2 \cong CP^1$, the spinor–null-direction correspondence of Penrose and Rindler, quoted from memory, not checked here against the source). The pair rule is the Born rule of a qubit written on light directions: a translation by the AI, not new.
- **T1, the coin of 2.40.** With $\chi \neq \pi$ the correlation depends on the sum of the angles, and C4 fails. With k = 1 one gets $−\cos 2\theta$ in the x–z plane, the photon curve, and isotropy in space breaks. With $p \neq 2$, C1 fails, with S = 3.469182 (p = 3) and $3.771236 = 4 \cdot (2\sqrt{2}/3)$ (p = 4). These weights are not quantum mechanics. So this does not violate the quantum Tsirelson bound.
- **A registered expectation did not come true.** For T4 with q = 2 (probability $(\eta/\eta_{opp})^2$) a failure of C1 was expected. Instead $E = f(−c) − f(c) = −c$ exactly. But the marginal of B is $(1 − c^2)/2$, up to 0.5 at $\theta = \pi/2$. So C2 fails.
- **Compatibility with the experiments.** No translation that holds deviates from $−\cos\theta$. It is a compatibility with the experiments, not a prediction. The values are read in the articles of the inventory (Sakai from the abstract only): Hensen and colleagues 2015, S = 2.42 ± 0.20; Storz and colleagues 2023, 2.0747 ± 0.0033; Poh and colleagues 2015, 2.82759 ± 0.00051; Sakai and colleagues 2006, proton pairs in the singlet, 2.83 ± 0.24 ± 0.07.

Reading for MM. The sentence of the common introduction on the mirror reciprocity of the two ends holds as a reason for the anticorrelation and for the form of the correlation. It does not hold for the curve: the curve comes from the law of one end. In every translation that holds, three ingredients are imported: the fair outcome of the first end read (v), the mirror as an update rule (iv), and the three directions as directions of the end, by isotropy or with the fourth side (iii).

Verification. Two AI verifiers: 11 remarks from the first, one of grade A (the protocol wrote the T4 rule as $\eta/2$, true only in normalised coordinates; the code uses the ratio, which is now the declared ingredient); 9 remarks of grade B/C from the second. Both: "ready with corrections". **No remark changed an outcome.**

5. Where MM stands among quantum interpretations

There is a classification called **Local Friendliness**. It says which assumptions must be given up by whoever accounts for quantum experiments. Cavalcanti and Wiseman (2021) define **Local Agency** like this: "the only relevant events correlated with an intervention are in its future light cone". Local Agency replaces Locality and No-Superdeterminism together. Then they split the relativistic causal arrow in two: the **temporal causal arrow** (causes come before effects) and **relativistic causality** (no influences faster than light).

	placement of August 2026 (status report 2.7–2.27)	**placement from 13 October 2026**
Absoluteness of Observed Events	kept in the minutes of 24 August; revised on 25 August ("the object is absolute, the reading partial")	**abandoned in the original definition.** A reading made inside a laboratory that lets no copies out (the friend in an extended Wigner scenario) is not absolute; readings with enough copies agree (revision of 25 August)
No-Superdeterminism	kept	**violated** by the dependence on future inputs, as in the model of 4.4 (not a conspiracy of initial conditions)
Local Agency	violated	violated (follows from the row above)
temporal causal arrow	kept	**abandoned** (the loop)
relativistic causality	abandoned ("realist non-	**kept as an assumption of the programme** (the author's choice, "variant A", 13 October

local")	2026). The pair model realises a local mediation along the paths; whether the cited principle is satisfied requires a separate causal argument. If causation were taken as transitive, the chain choice → reading → source → other reading would link space-like separated events

This is a placement, not a result. Two-time formulations that reproduce all the standard probabilities are equivalent to quantum mechanics in experiments, in that domain. For MM this equivalence is not yet shown.

6. Corrections to texts already published

Versions 2.7–2.18 of the status report contain the texts on the August placement. (Versions 2.17, 2.19 and 2.20 were not published.) In Italian they are there from 2.7; in English, as `QUANTUM_PLACEMENT.md`, from 2.9. The archive `MM_v8_addendum_26082026.zip` contains `PROPOSTA_scrivere_e_fondere.md`. Versions 2.21–2.27 carry these texts inside the inherited archive `MM_base_pubblicata_2.18.zip`. These statements are corrected:

1. **"The sense is the one Bell's theorem is about"** (QUANTUM_PLACEMENT §0). "Non-local" now means pregeometric (section 3). Bell correlations are carried by the loop, with local mediation along the paths (section 5).
2. **The conditional theorem** ("no fact determines which outcome is written") is withdrawn in the form in which it was written. Let us take two bits r and m, with a record that is blank at the start, m = 0. The reversible map U(r, q, 0) = (r, q, r) writes an antecedent relational fact into a record, without merging anything. The choice of the outcome is now a separate working assumption (W2), to be formalised.
3. **"The conservation axiom coincides with the purification axiom of Chiribella, D'Ariano and Perinotti"** (and the book "inherits their theorems"). It is a resemblance, to be shown formally. Their reconstruction also uses other axioms (PRA 84, 012311, 2011).
4. **"Three debts shown by others to be unavailable."** They are conditional constraints, not proofs of impossibility:
5. weight → frequency: Schlosshauer and Fine, Found. Phys. 35, 197 (2005), analyse the assumptions of the derivations;
6. non-locality: Bell and Local Friendliness constrain the explanations that keep certain assumptions;
7. a stable probability measure: the same family as weight → frequency.
8. **The table of the August placement** is replaced by section 5.

A correction to the book. The book NUMERI (v119, §[01435]) says that the Born exponent follows "from these premises alone". **That claim is withdrawn:** the argument also assumed a linear dynamics. Strong positivity gives a representation in a Hilbert space. But this work derives from it neither the dynamics that is needed nor the Born frequencies.

7. What is not claimed, and what remains open

Not claimed:

- a derivation of the Born rule without assumptions. In sections 4.1–4.5 quadratic weights, local Born responses and suitable measures over futures are assumed. Section 4.6 (2.48.1) obtains the Born rule from the counting of copies of equal modulus only with the declared assumptions H-es, H-eq and H-add (and H-cont for irrational ratios). The construction is similar to Zurek's, but H-eq is a declared premise. The physical necessity of H-eq is not derived;
- a derivation of the Tsirelson bound beyond the published theorems;
- a derivation of the singlet curve $-\cos\theta$ without imported ingredients. In section 4.7 (2.48.2) the curve holds in three translations, each with the ingredients listed there: law of one end, mirror update, fair first outcome, three directions;
- an experimental distinction from standard quantum mechanics;
- mathematical priority.

What remains open:

1. a rule that selects one outcome, with a measure over futures that gives Born. Section 4.5 shows that weights other than Born's make the consistent histories as improbable as one likes, in the pair model. To make this a reason for the Born weight, a postulate (written there) and a general argument are needed;
2. why the copy that returns is the complex conjugate of the one that leaves. In the model it follows from defining the return with the adjoint step (section 4.6); a physical reason remains open. And why copies of equal modulus count the same (H-eq, which in section 4.6 is the author's premise U1);

3. a locally mediated model beyond one singlet pair: more particles, networks, choices that adapt;
4. why the hidden parameter is not accessible, and a causal argument for relativistic causality;
5. the value of B(p) for $1 < p < 2$, and its dependence on the dimension.

How the work was done

Author, licences and assistants. Author: Ugo Lanciano, independent researcher. Original text and materials: CC BY 4.0. Original code: MIT licence. The calculations were made with AI assistants (Claude and Codex), and they were not reviewed by a human referee. Each block, and the archived text, was checked by an independent AI system (gpt-6-astra via Codex). Its reports and the changes that followed from them are kept in the working records. The protocol of block 06 was checked by gpt-6-luna via Codex. In block 07 the two verifiers are Astra (Codex, gpt-6-luna) and Argon (Gemini); the message to Argon is the same material sent to Astra.

Working folders. In the working records the material is in folders named 01_tsirelson, 02_copie_correlate, 03_loop_coerenza, 04_verbale_loop, 05_modello_loop, a second round R2 and R3. The working files are in Italian. The sections of the text correspond to the blocks as follows: 4.1 → 01_tsirelson; 4.2 → 02_copie_correlate; 4.3 → 03_loop_coerenza; 4.4 → 05_modello_loop; 4.5 → R3; 4.6 → 06_born_conteggio; 4.7 → 07_singoletto; 5 → 04_verbale_loop.

Registrations (13 October 2026, UTC times).

block	registration (UTC, 13 October 2026)	SHA-256 of the registered file
01 Tsirelson	15:41:53	6e0277aa…
02 two slits, first run	17:48:40	82a5526d…
02 two slits, second run	17:51:58	07922f3e…
03 loop consistency	18:08:51	683fcfba…
04 placement, author's choice of variant A	19:14:47	d252047d…
05 pair model	19:17:49	d7711e20…
R2 second round	19:30:20	b5c1550f…
R3 self-consistency	20:09:57	a9ecca20…

Block 06, new in 2.48.1 (UTC, 25–26 October 2026; registers in `documenti/registrations_original/06_born_conteggio/`). The author's premises with the exact words are in `PREMESSE_UGO_RACCOLTE_20261026.md`; the fingerprints of the 22 sources are in `IMPRONTE_FONTI.txt`; the program is `codice/06_born_conteggio/conti_K1_K5.py`.

step	registration (UTC)	SHA-256 of the registered file
author's premises and candidate sources	25 Oct 23:49:56	f1379750…, 7e8dcb0b…
fingerprints of the 22 sources	25 Oct 23:53:43	ac22ebb3…
Level-0 inventory and the four readings	26 Oct 00:10:09	9664c7e7… (inventory)
Level-1 protocol	00:11:30	2acc00b8…
first verification and corrections 1	00:31:55	be792a27… (protocol correction)
second verification, correction 2 and program	00:37:02	46c25746… (correction 2), 763b4d80…

		(program)
results of K1–K5	00:37:28	8fc342c1…
K5bis, added after the run, registered before running it	00:37:50	0150d11b…
outcome note	00:38:37	73ef4a06…

Block 07, new in 2.48.2 (UTC, 29 October 2026; registers in `documenti/registrations_original/07_singoletto/`). The program is `calcoli_livello1_c1.py`; the verifiers' corrections are in the outcome note v3 (`documenti/working_notes_it/07_singoletto_NOTA_ESITO_LIVELLO1_v3.md`).

step	registration (UTC)	SHA-256 of the registered file
author's premises and Level-0 inventory v0	13:01:10	69f84bb2…, f7d986c8…
fingerprints of the sources (7 on disk, 22 downloaded)	13:01:10, 13:04:32	ee53132d…, ed075ebf…
inventory v1, Level-1 protocol and program	13:24:30, 13:24:31	cb7ef867…, 76a0bcbb…, 164382b9…
correction 1 and corrected program	13:25:09	57dcd022…, 43fcd771…
outcome note v1	13:26:55	122795f3…
Astra G1 report and note; outcome note v2	13:32:22–13:33:20	30e4d6b9…, 94cfcd64…, 56c3e141…
Argon G1 note; outcome note v3	13:53:02	96489c61…, a660457a…

For block 07 the 29 source PDFs, their extracted text and the full messages to the verifiers are also left out. The 22 source PDFs of block 06 and their extracted text are also left out of the package (not redistributed; only the fingerprints remain), and so are the messages to the verifier, which contain extracts of the sources. The full fingerprints and the times are in `documenti/registrations_original/*/REGISTRO.txt`, next to the original prediction files, unchanged. A fingerprint with a local time documents the content; it does not by itself certify the time. The Italian working notes after the verification are in `documenti/working_notes_it/`.

How to redo the calculations.

- `python codice/run_all.py --short` runs everything except the three long programs (about 15 minutes).
- `python codice/run_all.py` runs everything (about 2 hours; `tsirelson.py` alone takes about 80 minutes).
- `python codice/run_all.py --compare-only` compares again an existing `uscite/` folder.

Block 06_born_conteggio (2.48.1) runs in both modes and takes a few seconds. Its reference log, `risultati/06_born_conteggio/log_K1_K5.txt`, is the registered log without the three timing lines (real, user, sys) added by the shell's `time` command, which are not program output. The registered log is kept unchanged in `documenti/registrations_original/06_born_conteggio/02_livello1/`.

Block 07_singoletto (2.48.2) runs in both modes and takes less than a minute. It runs `calcoli_livello1_c1.py`, the corrected program registered before its run (correction 1, section 4.7). The first registered version is kept in `documenti/registrations_original/07_singoletto/02_livello1/`.

Each block runs in `uscite/<block>/` and is compared with `risultati/<block>/` according to the rules written in `run_all.py`:

- standard output only, with the expected numpy division warnings removed from the references;
- running times masked only where they are running times (end-of-line parentheses, "FINE in", "calcolate in"), plus the start dates and the numpy version;
- decimal numbers equal within a relative tolerance of 10^{-9} plus an absolute one of 10^{-12}; integers (and numbers written as integers in the logs), booleans, infinities and NaN exactly equal;
- JSON compared without the time fields ("secondi", "sec"), right after each program and again at the end; figures

not compared;
- the exit code of each program stored; a missing output, a missing status or a non-zero exit is a failure;
- `controllo_d3.py` checked as a matching prefix with exactly 8 starts and a final line.

"Matching" means in agreement with the stored references according to these rules. It does not certify the references themselves. Requirements: Python 3, numpy, scipy, matplotlib (tested with Python 3.13, numpy 2.4.6, scipy 1.18.0).

The reference for `tsirelson.py --rapido` was regenerated with the final program. The working quick run had been made before the declared changes (faster test of the marginals, added seeds) and is kept as `risultati/01_tsirelson/log_rapido_codice_precedente.txt`. The registered outcomes come from the full run.

The programs are the working files, unchanged, with one exception: `controllo_d3.py` reads the number of starts from the environment variable PARTENZE, because the working run was stopped by hand after 8 starts.

Files of the package.

- `codice/<block>/`: the programs; `codice/run_all.py`
- `risultati/<block>/`: reference logs, JSON files and figures
- `documenti/REGISTERED_TESTS.md`: every registered criterion with its outcome
- `documenti/registrations_original/`: predictions and registers as they were written
- `documenti/working_notes_it/`: Italian notes after each round of verification
- `LEGGIMI_IT.md`: summary in Italian
- `codice/06_born_conteggio/`, `risultati/06_born_conteggio/`: block 06, new in 2.48.1
- `documenti/registrations_original/06_born_conteggio/`: the author's premises, the fingerprints of the sources, the readings, the inventory, the protocol, the corrections, the verification reports and the registers of block 06
- `documenti/working_notes_it/06_born_conteggio_NOTA_ESITO_LIVELLO1_v1.md`: Italian outcome note of block 06
- `IMPRONTE.sha256`: fingerprints of every file

This version. This student version was written with an AI assistant (Claude) from the archived text in English. First a complete Italian translation was made, then the rewriting for a school student. The numbers were checked with a program: the translation against the English text, this version against the translation. The archived text is the reference.

This English text is a translation of the Italian student version, which the author read and approved.

Glossary

- **Amplitude.** The number, like an arrow with a length and a direction, that quantum mechanics gives to each outcome.
- **Born rule.** The probability of an outcome is the square of the amplitude: $|\psi|^2$.
- **CHSH, score S.** A number that measures how much the results of two distant sides agree: at most 2 with common instructions, at most $2\sqrt{2}$ in quantum mechanics, at most 4 without signals.
- **Complex conjugate.** The same complex number with the sign of the imaginary part changed.
- **Feynman checkerboard (MM 2.40).** A model in which a particle moves forward in a zigzag; each turn has an imaginary weight.
- **Loop.** The assumption W2: the outcome written in a record is selected by the future end of its history, which closes on itself.
- **Marginals.** The averages of the results of one side only, looked at on its own. They are zero when every measurement of that side, on its own, gives +1 and −1 with the same frequency.
- **Non-local (in MM).** Not located in spacetime, pregeometric. It does not mean an influence faster than light.
- **Registered.** Written in a file and given a fingerprint before the test.
- **Retrocausal.** A model in which a cause in the future has an effect in the past.
- **Self-consistency.** The idea (Novikov) that only histories without contradictions can exist, even if the future acts on the past.
- **Signalling.** Being able to send a message from one side to the other. In known physics it cannot be done faster than light.
- **Singlet.** A state of two linked particles: measured in the same direction, their spins always give opposite results; their correlation is $-\cos\theta$.
- **Tsirelson bound.** The maximum of S in quantum mechanics: $2\sqrt{2}$, about 2.83.
- **TVD.** Total variation distance between two probability distributions: 0 if equal, 1 if completely different.

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